# M1/2A internship
## Dynamic cable simulation for ropeway transport systems

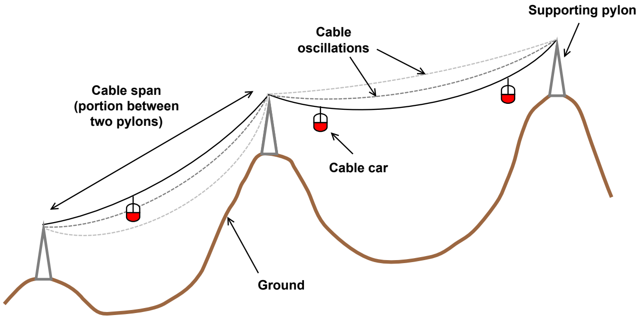


**Figure 1:** Left: cable car installation in French Alps, two supporting pylons are visible (source: Wikipedia)., Right: Schematic view of the installation, including the considered cable oscillations.

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**Keywords** Dynamical systems, Cable transports, Hamiltonian formalism, Numerical simulation

**Context** A cable transport system (cable car, chairlift, . . . ) is composed of one or several running cables, supported by several pylons, that carry cabins or other vehicles from one end to the other end of the cable path. Cable transport system are broadly used in ski resorts to transport skiers up mountains, but they are also more and more used in cities as an alternative to buses and tramways. The broad context of the internship is the dynamical simulation of cable transport systems subject to perturbations such as emergency brakes. When an emergency brake occurs, or when the installation restarts after a stop, oscillations may appear along the cable. These oscillations are often not due to the cable elasticity, but they correspond to a back-and-forth motion of the cable across a pylon, from one span to the next.

**Objectives** The goal of the internship is to formulate and simulate a simplified dynamical system, in order to study these nonlinear oscillations. In the simplified system, a cable supported by several pylons is reduced to a small-dimension dynamical system. The only variables are the lengths of cable between successive pylons. The student will

- Derive motion equations for the simplified system and study it theoretically
- Implement numerical simulations of the system using a high-level programming language
- Compare the simulation results with more accurate simulations to assess the impact of the simplification.
- Explore enriched versions of the system, adding cabins hanging to the cable, friction, etc.

The student should have a strong background in mechanics or applied mathematics, especially dynamical systems and numerical simulations. Experience in scientific programming (Python, Julia, . . . ) and knowledge of the Lagrangian/Hamiltonian formalism are a plus.

### Working environment

- The student will be hosted in the Tripop team at Centre Inria Grenoble Alpes, in Montbonnot, 30 minutes bus away from Grenoble.
- Supervised by Guillaume Mestdagh (postdoctoral researcher) and Vincent Acary (research director).
- Duration: 3 months (flexible), with gratification.
- How to apply: send your CV and transcript of records to guillaume.mestdagh@univ-grenoble-alpes.fr.

## References

[1] D Bruno and A Leonardi. Nonlinear structural models in cableway transport systems. *Simulation Practice and Theory*, 7(3):207–218, 1999.

[2] Herbert Goldstein, Charles Poole, and John Safko. *Classical mechanics*. Addison-Wesley, third edition, 2002.